

CoppeTeX quick reference

Class `ufrj`, v4.1 — every option, command and environment

This is the short companion to `ufrj.pdf`, the full manual, which is written in Portuguese. A UFRJ thesis may be written in Portuguese, English or Spanish (CEPG Resolution 302/2024, art. 57); this sheet is for whoever writes it in English. A thesis starts with the class and the style of its academic unit, `\documentclass[dsc]{ufrj}` and, at COPPE, `\usepackage{ufrj-coppe}`; everything below is the class.

The **Where** column says where a command may appear: PRE means the preamble, before `\begin{document}`; DOC means after `\begin{document}`, and, for everything that feeds the cover, *before* `\maketitle`.

Three things that surprise people. The document data goes *inside* document, not in the preamble. The bibliography is `biblatex` with `biber`, not BibTeX. And `\frontmatter`/`\mainmatter` are what switch folio numbering on, so leaving them out gives the wrong page numbers.

Command	Example	Where	What it does
CLASS OPTIONS — work type, one is required			
<code>msc</code>	<code>\documentclass[msc]{ufrj}</code>	PRE	M.Sc. dissertation.
<code>dsc</code>	<code>\documentclass[dsc]{ufrj}</code>	PRE	D.Sc. thesis.
<code>mscexam</code>	<code>\documentclass[mscexam]{ufrj}</code>	PRE	M.Sc. qualifying exam. No CAPES sheet, no catalogue card, no reference above the abstract.
<code>dscexam</code>	<code>\documentclass[dscexam]{ufrj}</code>	PRE	D.Sc. qualifying exam. Same.
<code>mscsem</code>	<code>\documentclass[mscsem]{ufrj}</code>	PRE	M.Sc. seminar, a requirement of some Programas and <i>not</i> a qualifying exam. Same three omissions: it is not deposited.
<code>grad</code>	<code>\documentclass[grad]{ufrj}</code>	PRE	Undergraduate work (TCC; “Projeto de Graduação” with <code>ufrj-poli</code>), presented to a Course: the title conferred is the course’s, and the additional sheet carries only the catalog card, without the CAPES fields.
CLASS OPTIONS — writing language			
<code>brazilian</code>	<code>[brazilian,dsc]</code>	PRE	Portuguese. The default; needs no writing.
<code>english</code>	<code>[english,dsc]</code>	PRE	English. The foreign abstract switches to Portuguese.
<code>spanish</code>	<code>[spanish,dsc]</code>	PRE	Spanish. Needs <code>ufrj-lang-spanish.def</code> and <code>spanish-ufrj.lbx</code> . Three abstracts.
<code>french</code>	<code>[french,dsc]</code>	PRE	French. Not a permitted thesis language; ships as a demonstration.
<code>italian</code>	<code>[italian,dsc]</code>	PRE	Italian. Same.
CLASS OPTIONS — the rest			
<code>numbers</code>	<code>[dsc,numbers]</code>	PRE	Numeric citations (1) instead of author-date (4.1.1.1.1: parentheses or superscript, <code>\supercite</code>); list in citation order. Not with footnotes: the class warns.

Command	Example	Where	What it does
sempdfa	[dsc,sempdfa]	PRE	Turns OFF the PDF/A-2b output, which is ON by default since v5.0: 2.2(d) requires the deposited file to be PDF/A (ISO 19005), self-contained — embedded fonts, sRGB colour, XMP metadata, nothing external. Use it for a draft run, or for a package PDF/A rejects; the deposited file has to be PDF/A. The v4.1 option pdfa is still accepted and does nothing. Check with veraPDF.
comserifa	[dsc,comserifa]	PRE	Serif family. Sans is the default since v4.1.
semserifa	--	PRE	Accepted, does nothing. What it asked for became the default.
semlinks	[dsc,semlinks]	PRE	Accepted; confirms the default (links with no colour and no frame, still clickable). Wins over the two below.
linkscmoldura	[dsc,linkscmoldura]	PRE	Framed links, on screen only (the frame is not printed).
setavermelha	[dsc,setavermelha]	PRE	Red curved arrow at the start of a wrapped listing line. Black by default.
linkscoloridos	[dsc,linkscoloridos]	PRE	Coloured link text. Against 2.1(b) of the UFRJ manual (black text): for on-screen reading copies, not for deposit.
doublespacing	[dsc,doublespacing]	PRE	Double spacing instead of one-and-a-half (2.2c).
twoside	[dsc,twoside]	PRE	Mirrored margins, for printing and binding.
oneside	[dsc,oneside]	PRE	One-sided, 3 cm left margin on every sheet. The default.
semorientadornabanca	[dsc,semorientadornabanca]	PRE	Approval sheet with the examiners only. By default the advisors head the board, as 3.1.2.1.3(e) requires.
orientadorexamina	[dsc,orientadorexamina]	PRE	Puts the advisors and coadvisors at the head of the board on the approval sheet. Off by default: the board is what examiner declares, in order, and the advisor, who chairs it, is the first examiner (3.1.2.1.3e).
coorientador	[dsc,coorientador]	PRE	List coadvisors on the abstract pages too.
rascunhoficha	[dsc,rascunhoficha]	PRE	Draft catalogue card while writing. Never valid for deposit.
resumosemreferencia	[dsc,resumosemreferencia]	PRE	Drop the work's own reference from the top of the three abstracts. It is printed by default (3.1.2.1.4, Annexes E and F).
listasnosumario	[dsc,listasnosumario]	PRE	Put the pre-textual lists in the table of contents. Against NBR 6027 and the UFRJ manual (3.1.2.1.6): use only if your Programa requires it.
simbolosalfabeticos	[dsc,simbolosalfabeticos]	PRE	List of symbols in alphabetical order instead of order of appearance.

Command	Example	Where	What it does
<code>assinaturas</code>	--	PRE	Accepted, does nothing. Signature rules are gone: deposit is digital.
<code>numeraisromanos</code>	<code>[dsc,numeraisromanos]</code>	PRE	Roman numerals on pre-textual sheets. Against 2.7 ; legacy only.
<code>morewrites</code>	<code>[dsc,morewrites]</code>	PRE	A missing <code>morewrites</code> package becomes an error, not a warning.
<code>semmakeindex</code>	<code>[dsc,semmakeindex]</code>	PRE	Do not run <code>makeindex</code> from the class. By default the class runs it at the end of each compilation (restricted shell escape is enough), so the lists and the index appear in any editor.
<code>semmorewrites</code>	<code>[dsc,semmorewrites]</code>	PRE	Do not load <code>morewrites</code> . Saves about half a second per <code>pdflatex</code> pass (Overleaf free plan). Safe only without an index, several indexes, a glossary or nomenclature: the class stops with a clear error if the 16 streams run out.
PREAMBLE			
<code>\addbibresource</code>	<code>\addbibresource{thesis.bib}</code>	PRE	Declare the bibliography database. With the <code>.bib</code> extension.
<code>\makelosymbols</code>	<code>\makelosymbols</code>	PRE	Turn on collection of the list of symbols. Without it, <code>\syml</code> records nothing.
<code>\makeloabbreviations</code>	<code>\makeloabbreviations</code>	PRE	Same for the list of abbreviations.
<code>\makelosiglas</code>	<code>\makelosiglas</code>	PRE	Separate list of acronyms (recommended by 3.1.2.2.6). Without it, acronyms go to the list of abbreviations.
<code>\newufrjfloat</code>	<code>\newufrjfloat{map}{Map}{List of Maps}</code>	PRE	Declare your own float: caption on top, <code>\source</code> below, own list.
<code>\newsigla</code>	<code>\newsigla{ufrj}{UFRJ}{Universidade Federal...}</code>	PRE	Declare an acronym for <code>\sigla</code> .
<code>\useufrjlanguage</code>	<code>\useufrjlanguage{french}</code>	PRE	Load one more language pack, for quotations in a third language.
<code>\graphicspath</code>	<code>\graphicspath{{figures/}}</code>	PRE	Where your figures live. Standard L ^A T _E X.
<code>\freeconfig</code>	<code>\freeconfig{Dr.}{...}{...}{...}{...}{...}</code>	PRE	Override the work type entirely. Escape hatch; do not use for a normal thesis.
IDENTIFICATION — after <code>\begin{document}</code>, before <code>\maketitle</code>			
<code>\title</code>	<code>\title{Título da tese}</code>	DOC	The Portuguese title.
<code>\foreigntitle</code>	<code>\foreigntitle{Thesis title}</code>	DOC	The English title.
<code>\titlein</code>	<code>\titlein{spanish}{Título}</code>	DOC	The title in any other language.
<code>\subtitle</code>	<code>\subtitle{Um subtítulo}</code>	DOC	Optional subtitle, Portuguese. Printed after a colon, in ordinary case.
<code>\foreignsubtitle</code>	<code>\foreignsubtitle{A subtitle}</code>	DOC	The same, English.
<code>\subtitlein</code>	<code>\subtitlein{spanish}{...}</code>	DOC	The same, any language.
<code>\author</code>	<code>\author{All given names}{Surname}</code>	DOC	Two arguments: the catalogue card has to invert the name.

Command	Example	Where	What it does
<code>\advisor</code>	<code>\advisor{Ana}{Lima}{D.Sc.}{UFRJ}</code>	DOC	Name, surname, degree, institution. Repeat per advisor. Optional first argument is the treatment.
<code>\coadvisor</code>	<code>\coadvisor{Bia}{Sousa}{Ph.D.}{UFF}</code>	DOC	Same four arguments.
<code>\examiner</code>	<code>\examiner{Carlos Dias}{D.Sc.}{USP}</code>	DOC	Name and surname in one argument, then degree and institution.
<code>\department</code>	<code>\department{PESC}</code>	DOC	The Programa's acronym, as the unit style declares it (<code>\ufrj-coppe</code> declares COPPE's thirteen).
<code>\date</code>	<code>\date{09}{2026}</code>	DOC	Month and year of deposit, as numbers. Not the defence date.
<code>\dataaprovacao</code>	<code>\dataaprovacao{15 de setembro de 2026}</code>	DOC	Defence date, in words. Without it the sheet reads "a ser determinada".
<code>\keyword</code>	<code>\keyword{Formal methods}</code>	DOC	One call per keyword, main language.
<code>\foreignkeyword</code>	<code>\foreignkeyword{Métodos formais}</code>	DOC	One call per keyword, foreign language.
<code>\braziliankeyword</code>	<code>\braziliankeyword{Métodos formais}</code>	DOC	Portuguese keywords when the work is not in Portuguese.
<code>\volumes</code>	<code>\volumes{3}</code>	DOC	How many volumes. Nothing is printed when there is one.
<code>\volumefiles</code>	<code>\volumefiles{v1,v2}</code>	PRE	The .tex names of every volume, in order, in the preamble of all of them: the sheet count runs from the first volume to the last (2.7) and every volume carries the complete sumario (3.1.2.1.6). Compile them alternately, twice. Alias: <code>\arquivosdosvolumes</code> .
<code>\volume</code>	<code>\volume{1}</code>	DOC	Which volume this is.
TITLE PAGE AND CAPES SHEET			
<code>\areaconcentracao</code>	<code>\areaconcentracao{Engenharia de Sistemas}</code>	DOC	Area of concentration. On the folha adicional; on the identity sheets only under the <code>areanafolhaderosto</code> option.
<code>\linhapesquisa</code>	<code>\linhapesquisa{Engenharia de Dados}</code>	DOC	Research line, on the title page. Label follows the main language. Empty is fine.
<code>\tipoproducao</code>	<code>\tipoproducao{bibliografica}</code>	DOC	One of <code>bibliografica</code> , <code>artistica</code> , <code>tecnologica</code> , <code>tecnica</code> .
<code>\projetovinculado</code>	<code>\projetovinculado{sim}</code>	DOC	<code>sim</code> or <code>nao</code> : is there a linked research project?
<code>\nomeprojeto</code>	<code>\nomeprojeto{Nome do projeto}</code>	DOC	Its name, if there is one.
<code>\agenciafomento</code>	<code>\agenciafomento{Conselho...}{CNPq}</code>	DOC	Funding agency, full name and acronym. One call each.
<code>\fichacatalografica</code>	<code>\fichacatalografica{ficha.pdf}</code>	DOC	The official catalogue card, as a PDF from the SiBI generator.
STRUCTURE			
<code>\maketitle</code>	<code>\maketitle</code>	DOC	Typesets cover, title page and CAPES sheet. After all the data.
<code>\frontmatter</code>	<code>\frontmatter</code>	DOC	Opens the pre-textual part and typesets the approval sheet.
<code>\mainmatter</code>	<code>\mainmatter</code>	DOC	Opens the textual part. This is what starts printing the folio.
<code>\backmatter</code>	<code>\backmatter</code>	DOC	Opens the post-textual part.
<code>\dedication</code>	<code>\dedication{To those who waited.}</code>	DOC	Dedication, on a sheet of its own.

Command	Example	Where	What it does
<code>\epigrafe</code>	<code>\epigrafe{quotation}{author}</code>	DOC	Epigraph (3.1.2.2.3), on a sheet of its own, after the acknowledgements.
<code>abstract</code>	<code>\begin{abstract} ...</code>	DOC	Abstract in the main language. One sheet, one paragraph, 250 words.
<code>foreignabstract</code>	<code>\begin{foreignabstract} ...</code>	DOC	Abstract in the foreign language.
<code>brazilianabstract</code>	<code>\begin{brazilianabstract} ...</code>	DOC	Third abstract, Portuguese, for a Spanish- or French-main work.
<code>\configuraresumos</code>	<code>\configuraresumos{true}{true}{true}{true}</code>	PRE	Turns the four optional elements of every abstract sheet on or off, in every language: page identification, title and author, advising, reference. By default only the reference, as in Annexes E and F; only the abstract text itself cannot be turned off.
<code>\appendix</code>	<code>\appendix</code>	DOC	Opens the appendices: text <i>you</i> wrote.
<code>\annex</code>	<code>\annex</code>	DOC	Opens the annexes: documents you did <i>not</i> write.
<code>\printbibliography</code>	<code>\printbibliography</code>	DOC	The reference list, titled “Referências” (3.1.4.1).
<code>\ufrjfinalpage</code>	<code>\ufrjfinalpage</code>	DOC	Optional colophon, every item measured at compile time: class version, engine and its version, TeX system, L ^A T _E X format, font family and encoding, biblatex version, date, hour, operating system and machine.

Only the table of contents is mandatory (3.1.2.1.6). Every list is optional (3.1.2.2); the COPPE norm (§10) adds the lists of *quadros*, programs and algorithms whenever the work has one of them.

PRE-TEXTUAL LISTS — illustrations, tables, abbreviations, symbols; contents last (3.1.2)

<code>\listoffigures</code>	<code>\listoffigures</code>	DOC	List of figures.
<code>\listofquadros</code>	<code>\listofquadros</code>	DOC	List of <i>quadros</i> (framed on all sides, unlike a table).
<code>\listofprogramas</code>	<code>\listofprogramas</code>	DOC	List of code listings.
<code>\listofalgorithms</code>	<code>\listofalgorithms</code>	DOC	List of algorithms.
<code>\listoftables</code>	<code>\listoftables</code>	DOC	List of tables, after every list of illustrations (3.1.2).
<code>\printloabbreviations</code>	<code>\printloabbreviations</code>	DOC	List of abbreviations and acronyms. No page numbers (4.1.1).
<code>\printlosiglas</code>	<code>\printlosiglas</code>	DOC	List of acronyms, when <code>\makelosiglas</code> is on.
<code>\printlosymbols</code>	<code>\printlosymbols</code>	DOC	List of symbols, in order of first appearance (3.1.2.2.7).
<code>\makeglossarylist</code>	<code>\makeglossarylist</code>	PRE	The automatic glossary (3.1.4.2). Without it, write <code>theglossary</code> by hand.
<code>\glossaryterm</code>	<code>\glossaryterm[key]{term}{definition}</code>	DOC	Registers a glossary term where it appears; prints nothing. Alphabetical, accents included.
<code>\printglossarylist</code>	<code>\printglossarylist</code>	DOC	The glossary, right after <code>\printbibliography</code> .
<code>\index</code>	<code>\index{term!sub see{other}}</code>	DOC	Index entry (with <code>makeidx</code>). Accented entries get an unaccented sort key; <code>key@term</code> wins.
<code>\acron</code>	<code>\acron[key]{UFRJ}{meaning}</code>	DOC	Registers an acronym; <code>\sigla</code> uses it.

Command	Example	Where	What it does
<code>\tableofcontents</code>	<code>\tableofcontents</code>	DOC	The table of contents. Comes last of all the lists.
ILLUSTRATIONS — caption on top, source below (2.10)			
<code>\source</code>	<code>\source{Prepared by the author.}</code>	DOC	The source line, below the float. Mandatory by the norm.
<code>\illustrationwidth</code>	<code>\illustrationwidth{4cm}</code>	DOC	Inside a float: sets the caption and the source line to the width of the illustration (2.10). Alias: <code>\largurailustracao</code> .
<code>\fonte</code>	<code>\fonte{Elaboração própria.}</code>	DOC	The same command, Portuguese name.
<code>\cpsource</code>	<code>\cpsource{...}</code>	DOC	Clash-safe alias, always available.
<code>\cpfonte</code>	<code>\cpfonte{...}</code>	DOC	Likewise.
<code>quadro</code>	<code>\begin{quadro}[ht] ...</code>	DOC	A float framed on all sides. Behaves like figure .
<code>python</code>	<code>\begin{python}[caption={...}]</code>	DOC	Python listing, syntax-highlighted, caption on top.
<code>java, xml, html, prolog</code>	<code>\begin{java}[...] ...</code>	DOC	The same, for those languages.
<code>\pythoninline</code>	<code>\pythoninline{f(x)}</code>	DOC	Code inside a paragraph.
<code>\javainline</code>	<code>\javainline{new X()}</code>	DOC	Likewise.
<code>\pythonexternal</code>	<code>\pythonexternal{code.py}</code>	DOC	Listing read from a file.
<code>\javaexternal</code>	<code>\javaexternal{Code.java}</code>	DOC	Likewise.
<code>\pythonstyle</code>	<code>\pythonstyle</code>	DOC	Apply a language's style without opening the environment. Also <code>\javastyle</code> , <code>\xmlstyle</code> , <code>\htmlstyle</code> , <code>\prologstyle</code> .
QUOTATIONS, ACRONYMS, SYMBOLS			
<code>\citacao</code>	<code>\citacao{up to three lines}</code>	DOC	Short quotation, inside the paragraph, in quotation marks.
<code>longquote</code>	<code>\begin{longquote} ...</code>	DOC	Quotation over three lines: indented 4 cm past the margin, smaller, single spaced, no quotation marks (2.2b).
<code>\abbrev</code>	<code>\abbrev[IOT]{IoT}{Internet of Things}</code>	DOC	Register an acronym. Optional first argument is the sort key.
<code>\syml</code>	<code>\syml{\${\alpha\$}}{Learning rate}</code>	DOC	Register a symbol. Same optional sort key.
<code>\sigla</code>	<code>\sigla{ufrj}</code>	DOC	Full form on first use, acronym afterwards (2.8). <code>\sigla*</code> forces the full form.
CITATIONS — biblatex, with the natbib layer on			
<code>\cite</code>	<code>\cite{key}</code>	DOC	(SURNAME, 2026), or (1) with numbers .
<code>\citet</code>	<code>\citet{key}</code>	DOC	SURNAME (2026).
<code>\citep</code>	<code>\citep{key}</code>	DOC	(SURNAME, 2026).
<code>\textcite</code>	<code>\textcite{key}</code>	DOC	Same as <code>\citet</code> , biblatex spelling.
<code>\parencite</code>	<code>\parencite{key}</code>	DOC	Same as <code>\citep</code> .
<code>\citeauthor</code>	<code>\citeauthor{key}</code>	DOC	The author alone.
<code>\citeyear</code>	<code>\citeyear{key}</code>	DOC	The year alone.
<code>\citetitle</code>	<code>\citetitle{key}</code>	DOC	The title alone.
<code>\cite[p. 45]</code>	<code>\cite[p. 45, our translation]{key}</code>	DOC	Page, and any note, in the optional argument.

Command	Example	Where	What it does
<code>\citepapud</code>	<code>\citepapud[45]{Original}{1953}{key-read}</code>	DOC	Citation of a citation (4.1.2.2.1). Three mandatory arguments: the original author and year, which are printed as text, and the key of the work you actually read, which is the only one that enters the list. Also <code>\citetapud</code> .
<code>\nocite</code>	<code>\nocite{key}</code>	DOC	Into the list without citing in the text.

Full manual: `ufrj.pdf`, in Portuguese. Worked examples: `min-exemplo.tex` (only what is mandatory) and `max-exemplo.tex` (everything). Problems: <https://github.com/COPPE-UFRJ/CoppeTeX/issues>